

Sri Sivasubramaniya Nadar College of Engineering, Chennai

Department of Computer Science and Engineering

ICS1502 - Introduction to Machine Learning CAT Assignment 1 - Report

Field	Details
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1. Aim

To (i) evaluate matrix-based linear regression (closed form and gradient descent, with/without L2 regularization and standardization) for mobile-phone price prediction, and (ii) fit a single-layer (logistic) classifier for bank-note authentication, studying L2 regularization, a 3-D decision surface and the effect of outliers.

2. Part 1 - Regression: Mobile Phone Price Prediction

2.1 Dataset and matrix formulation

The dataset has 161 phones and 12 numeric predictors (Sale, weight, resolution, ppi, cpu core, cpu freq, internal mem, ram, RearCam, Front_Cam, battery, thickness); the target is **Price**. Identifier columns were dropped and rows with missing values removed.

Data matrix X is 128×13 (training rows $\times$ (features + 1)); the leading column of ones lets the intercept be learned inside θ . **Label matrix y** is 128×1 (one real-valued price per phone, hence regression). **Parameter matrix θ** is 13×1 , fixed by the dimension of the data so that $X\theta$ is conformable; each θ_j is the change in predicted price for a one-unit change in feature j with the others fixed, so its sign and magnitude show the feature's influence on the prediction.

Only training rows are used to build X and y ; the test rows (33 phones, 20%) are used only for evaluation, giving an unbiased estimate of generalization. Conditioning: $\text{cond}(X^T X)$ is $2.54e+09$ for raw features versus $2.44e+00$ after standardization, which is why gradient descent is run on standardized data.

2.2 Effect of the train/test split

test_size	train_R2	test_R2_mean	test_R2_std	test_RMSE_mean
0.1000	0.7954	0.7308	0.1049	2,738
0.2000	0.7973	0.7399	0.0710	2,950
0.3000	0.8003	0.7441	0.0513	2,969
0.4000	0.8033	0.7439	0.0344	2,949
0.5000	0.8119	0.7327	0.0364	3,012

Over 30 random splits per ratio, mean test R^2 is highest at a test fraction of 0.3 (0.744), and the spread of test R^2 is largest at 0.1 ($\sigma = 0.105$). A small test set gives a noisy error estimate, while a large test set leaves less data for learning θ . The 80/20 split is a standard compromise and is used

throughout.

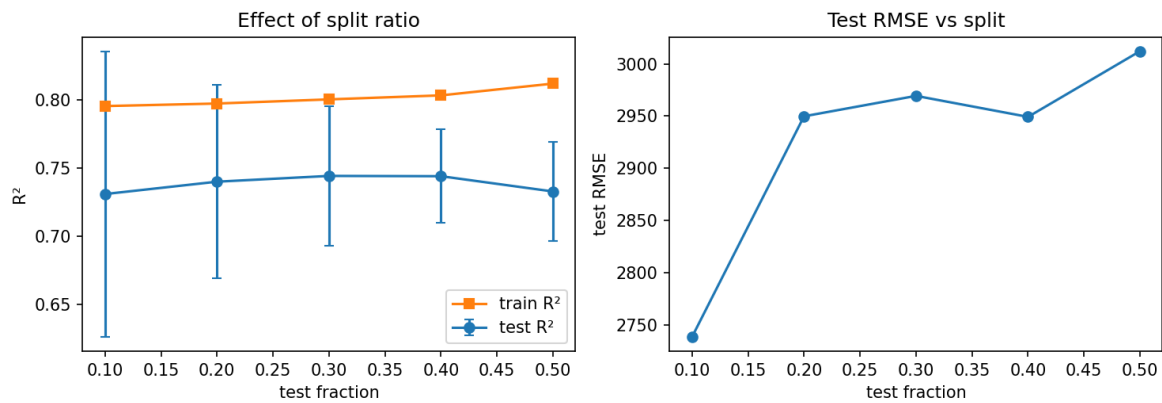


Figure 1: effect of the split ratio

2.3 Q1-Q3: closed-form vs gradient-descent solution

Closed form: $\theta = (X^T X)^{-1} X^T y$ (pseudo-inverse for numerical safety). Gradient descent: $\theta \leftarrow \theta - \alpha \cdot (1/n) X^T (X\theta - y)$ on standardized features and target, converted back to price units.

index	MSE	RMSE	MAE	R2	MAPE_%
Closed form (train)	6,129,344	2,476	1,998	0.8202	9.2587
Closed form (test)	12,730,571	3,568	2,671	0.6198	12.0330
GD (train)	6,129,344	2,476	1,998	0.8202	9.2587
GD (test)	12,730,540	3,568	2,671	0.6198	12.0329

GD converged in 327 iterations; the largest difference between the two methods' test predictions is 0.02645, i.e. both reach the same optimum of the convex least-squares loss. On the test set $R^2 = 0.620$, RMSE = 3,568 and MAE = 2,671; training R^2 is 0.820, so the train-test gap of 0.200 indicates some over-fitting/variance.

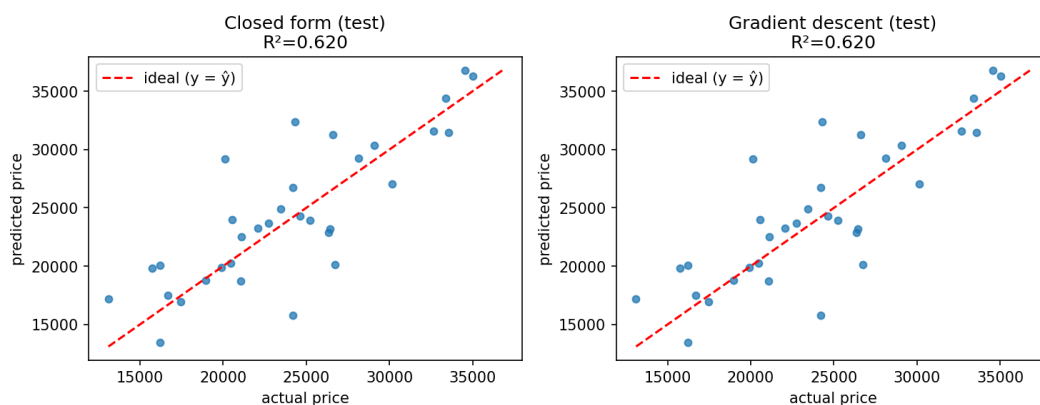


Figure 2: predicted vs actual (test) - closed form and gradient descent

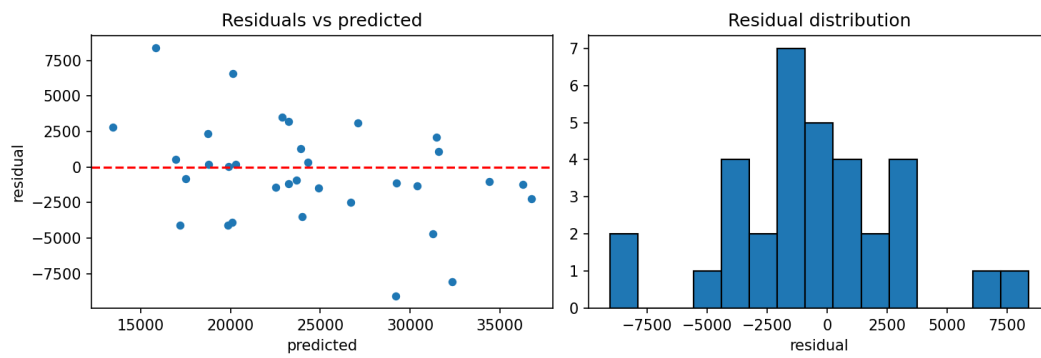


Figure 3: residual analysis (closed form)

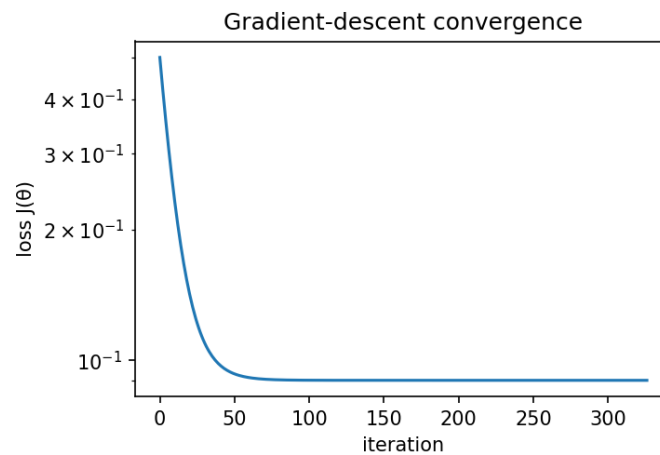


Figure 4: gradient-descent convergence

Error analysis: points close to the red $y = \hat{y}$ line are accurate; the residual plots show whether errors are centred on zero and whether they grow with price (funnel shape) – a sign that a purely linear model only approximates the price structure (brand, launch date and interactions are not captured).

2.4 Q4: L2 (ridge) regularization

Ridge solves $\theta = (X^T X + \lambda I)^{-1} X^T y$ (intercept not penalized); the same penalty is added to the GD gradient. λ was chosen by 5-fold cross-validation on the training set only: $\lambda^* = 3.162$.

index	MSE	RMSE	MAE	R2	MAPE_%
OLS closed (test)	12,730,571	3,568	2,671	0.6198	12.0330
OLS GD (test)	12,730,540	3,568	2,671	0.6198	12.0329
Ridge closed (test)	12,332,539	3,512	2,616	0.6317	11.8421
Ridge GD (test)	12,332,539	3,512	2,616	0.6317	11.8421
Ridge closed (train)	6,145,909	2,479	2,012	0.8197	9.3456
Ridge GD (train)	6,145,909	2,479	2,012	0.8197	9.3456

Ridge closed form and ridge GD agree (max test-prediction difference 0.00013). Compared with OLS, ridge test RMSE changes from 3,568 to 3,512 (+1.58% improvement), test R^2 from 0.620 to 0.632. Shrinkage reduced variance and gave slightly better generalization.

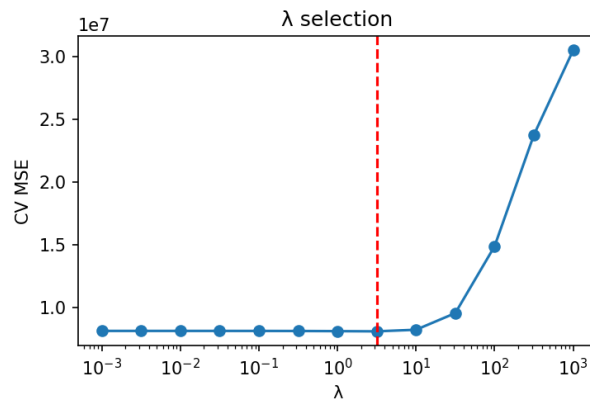


Figure 5: cross-validated MSE vs λ

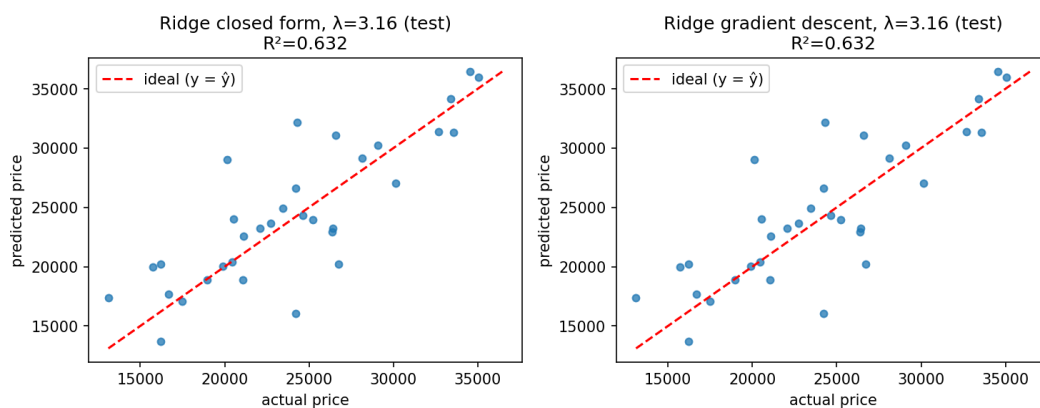


Figure 6: ridge predicted vs actual (test)

2.5 Q5: ridge with and without standardization

index	MSE	RMSE	MAE	R2	MAPE_%
Ridge closed form - RAW (test)	12,647,245	3,556	2,663	0.6223	11.9976
Ridge closed form - STANDARDIZED (test)	12,332,539	3,512	2,616	0.6317	11.8421
Ridge GD - RAW (test)	21,807,967	4,670	3,558	0.3487	15.7302
Ridge GD - STANDARDIZED (test)	12,332,539	3,512	2,616	0.6317	11.8421

With the same λ , ridge on raw features reaches test $R^2 = 0.622$ versus 0.632 when standardized. Without standardization the L2 penalty is unfair: features measured in large units (e.g. battery) need only small weights and are barely penalized, while small-scale features (e.g. CPU frequency) need large weights and are heavily shrunk. Gradient descent is affected even more: on raw data it used 20000 iterations and ended with $R^2 = 0.349$, whereas on standardized data it converged in 457 iterations with $R^2 = 0.632$. Standardization (statistics from the training set only, to avoid leakage) therefore gives a fair penalty, better conditioning and much faster, reliable convergence.

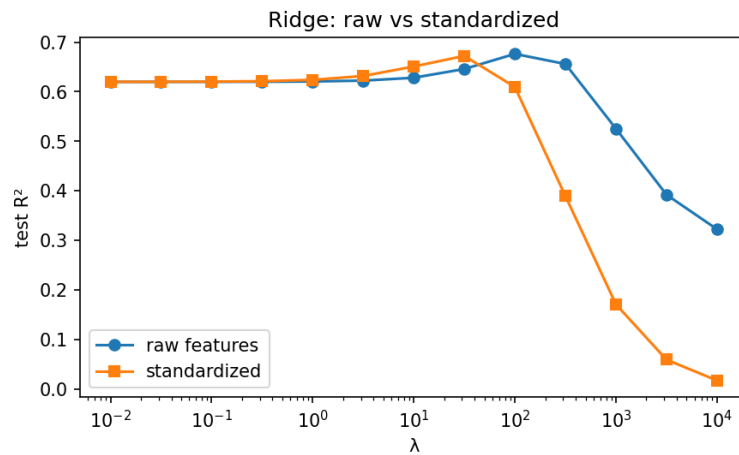


Figure 7: test R^2 vs λ , raw vs standardized features

2.6 Q6: effect of λ

lambda	test_R2	test_RMSE	train_R2	theta
0.0000	0.6198	3,568	0.8202	0.9013
0.1000	0.6202	3,566	0.8202	0.9006
1.0000	0.6238	3,549	0.8201	0.8941
10.0000	0.6508	3,420	0.8158	0.8343
100.0000	0.6095	3,616	0.6652	0.5048
1,000	0.1706	5,270	0.1824	0.1045
10,000	0.0168	5,738	0.0217	0.0117
100,000	-0.0019	5,792	0.0022	0.0012

Test R^2 is best at $\lambda = 10$ ($R^2 = 0.651$). Small $\lambda \approx$ OLS (low bias, high variance); a moderate λ gives the best bias-variance trade-off; for very large λ all weights shrink to ≈ 0 ($\|\theta\|$ falls from 0.901 to 0.0012) and every prediction collapses to the mean price ($R^2 \rightarrow -0.002$) – clear under-fitting, visible as a flat horizontal line of predictions.

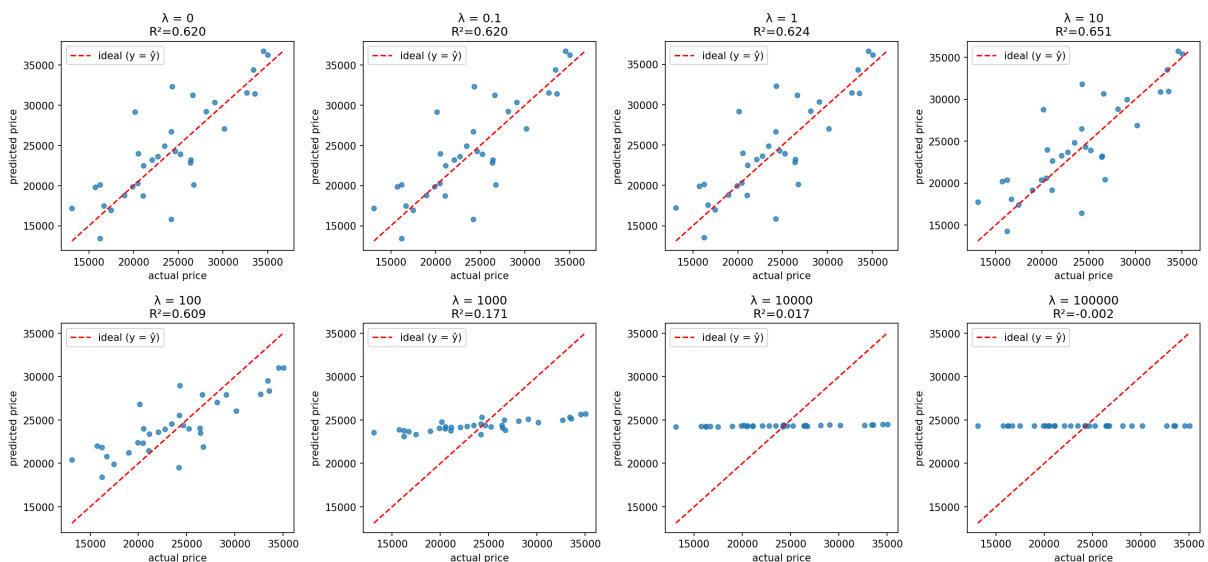


Figure 8: predicted vs actual (test) for different λ

2.7 Q7: feature importance from the L2 weights

rank	feature	ridge_weight	OLS_weight
1	ram	0.7661	0.7860

rank	feature	ridge_weight	OLS_weight
2	internal mem	0.3939	0.4038
3	thickness	-0.0899	-0.0935
4	battery	0.0755	0.0768
5	resolution	0.0665	0.0663
6	weight	-0.0602	-0.0588
7	RearCam	0.0557	0.0575
8	cpu freq	0.0525	0.0534
9	cpu core	-0.0479	-0.0488
10	ppi	0.0165	0.0166
11	Sale	-0.0163	-0.0161
12	Front_Cam	-0.0047	-0.0027

Because features are standardized, each weight is the price change per 1 σ change in that feature and weights are directly comparable. The most influential features are **ram** (raises price, $w = 0.766$); **internal mem** (raises price, $w = 0.394$); **thickness** (lowers price, $w = -0.090$). Ridge shrinks weak or collinear features towards zero (smallest: Front_Cam, $w = -0.005$), so large surviving weights mark real predictors. Caveat: correlated features share weight, so a small weight does not always mean irrelevance.

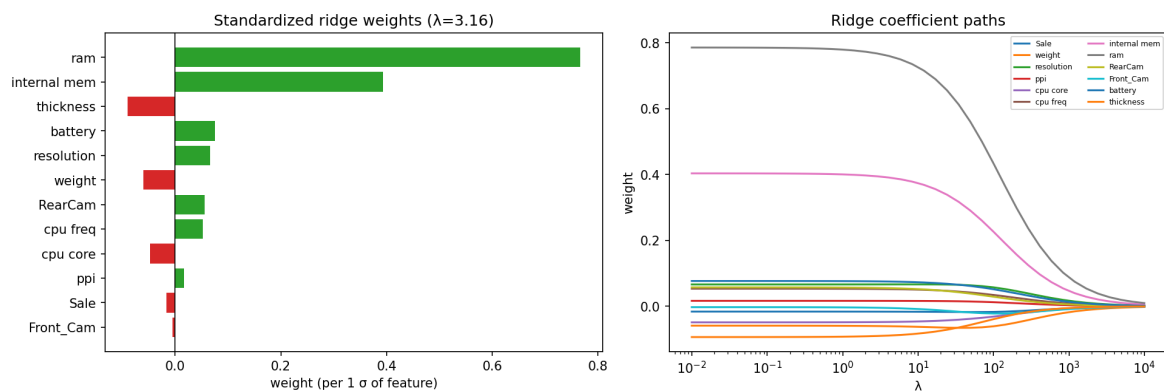


Figure 9: ridge weights and coefficient paths

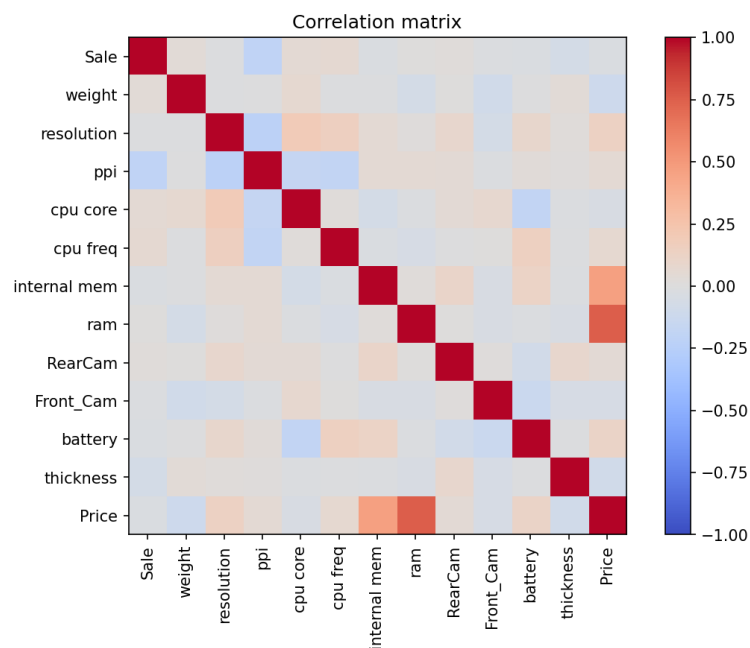


Figure 10: feature correlation matrix

2.8 Effectiveness of the matrix-based approach

The normal-equation/matrix approach gives an interpretable, fast model (test $R^2 \approx 0.62$, MAPE $\approx 12.0\%$). Strengths: exact one-step solution, transparent weights. Limitations: assumes linearity, is sensitive to outliers and multicollinearity, and cannot capture interactions or non-linear price effects; polynomial features or tree ensembles would be natural improvements.

3. Part 2 - Linear Classification: Bank Note Authentication

3.1 Model and data split

The dataset has 1372 notes with four wavelet features (variance, skewness, kurtosis, entropy) and a binary label (class 1 fraction 0.445). The model is a **single-layer neural network (logistic regression)**, $p(y=1|x) = \sigma(w^T x + b)$, trained by gradient descent on binary cross-entropy with optional L2 penalty $(\lambda/2n)\|w\|^2$. It is suitable because the task is two-class, the features are numeric and the classes are close to linearly separable.

Set	Samples	Class-1 fraction
Train	1097	0.4448
Test	275	0.4436

A stratified 80/20 split keeps the class ratio equal in both sets; features are standardized with training statistics only.

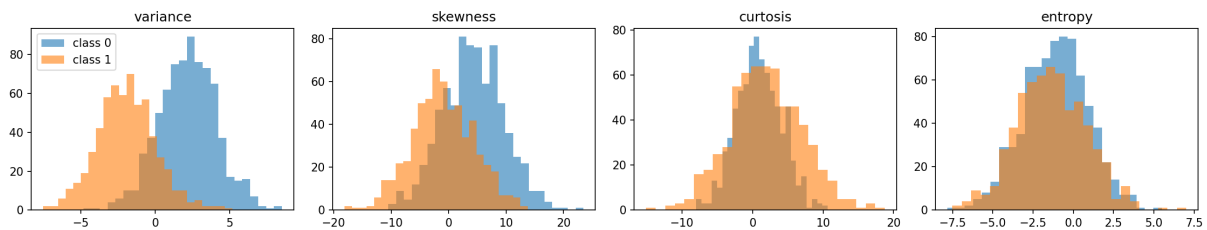


Figure 11: feature distributions by class

3.2 Q2: with and without L2 regularization

index	accuracy	precision	recall	F1	TN	FP	FN	TP
No L2 ($\lambda=0$) - train	0.8833	0.8659	0.8730	0.8694	543.0000	66.0000	62.0000	426.0000
No L2 ($\lambda=0$) - test	0.9091	0.9076	0.8852	0.8963	142.0000	11.0000	14.0000	108.0000
L2 ($\lambda=100$) - train	0.8906	0.8898	0.8607	0.8750	557.0000	52.0000	68.0000	420.0000
L2 ($\lambda=100$) - test	0.9164	0.9381	0.8689	0.9021	146.0000	7.0000	16.0000	106.0000

λ was chosen by 5-fold CV on the training set ($\lambda^* = 100$). Test accuracy is 0.9091 without L2 and 0.9164 with L2; the weight norm falls from 3.343 to 1.317. L2 regularization gave a small gain in test accuracy while keeping the weights smaller.

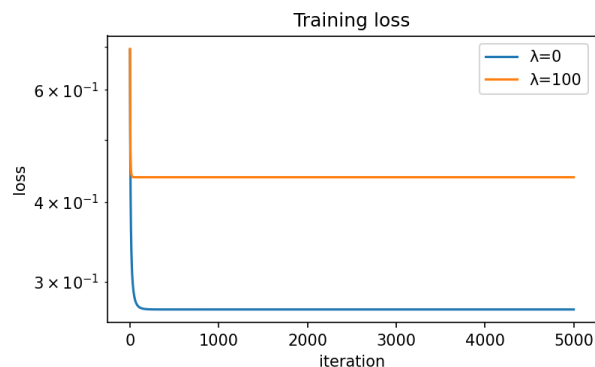


Figure 12: training loss

3.3 Q3: training and test accuracy vs λ

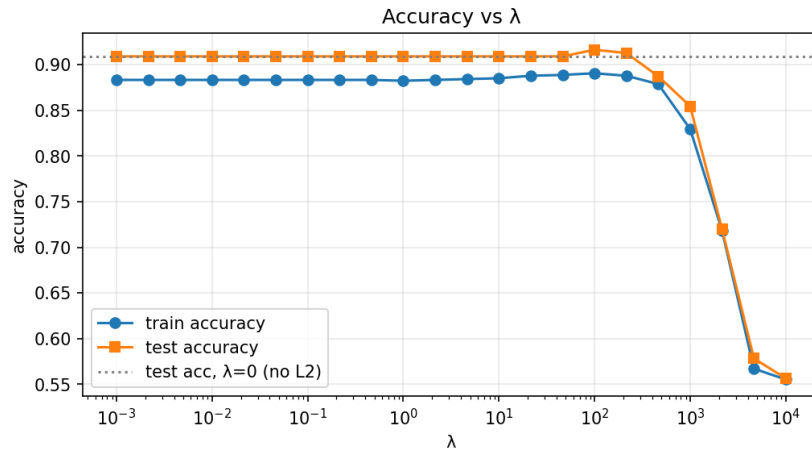


Figure 13: accuracy vs λ

Accuracy is flat for small λ , reaches its best test value of 0.9164 at $\lambda \approx 100$, and falls to 0.556 at $\lambda = 10000$ as the weights are over-shrunk (under-fitting). Train and test accuracy stay close together, so over-fitting is mild for this low-dimensional problem.

3.4 Q4: 3-D visualization of the three most important features

Ranked by $|w|$ in the standardized L2 model: variance (1.169), skewness (0.587), kurtosis (0.128), entropy (0.088). The top three – **variance**, **skewness**, **kurtosis** – were used for the 3-D plot. A classifier restricted to these three features still reaches 0.9127 test accuracy; the grey surface is its decision plane $w^T x + b = 0$ separating the two classes.

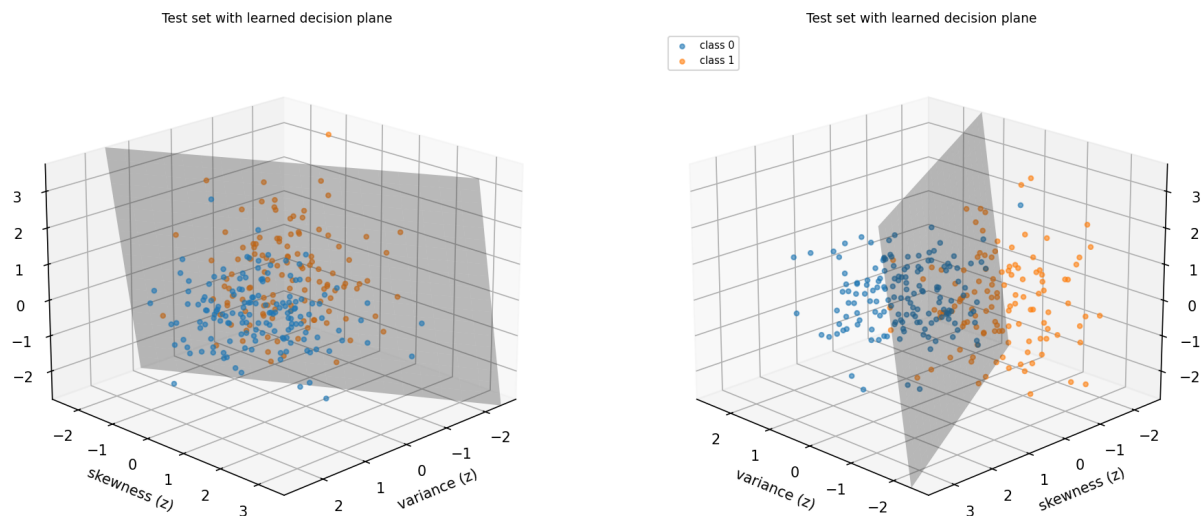


Figure 14: test data in 3-D with the learned decision plane

3.5 Q5: introducing outliers

A fraction of 5% of the training points (55 of 1097) was shifted by ± 6 standard deviations along variance, skewness, kurtosis (random sign, labels unchanged). The standardizer was re-fitted on the contaminated training data, and models were tested on the clean test set.

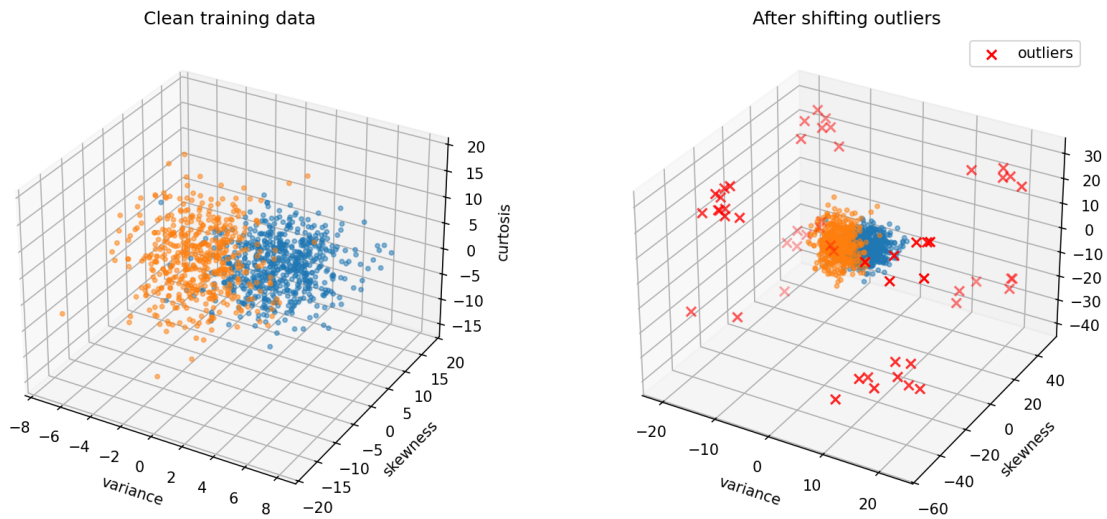


Figure 15: training data before and after shifting points

3.6 Q6: classifier on outlier-injected data

training data	λ	train_acc	clean_test_acc	$ w $
clean	0.0000	0.8833	0.9091	3.3427
clean	100.0000	0.8906	0.9164	1.3174
5% outliers	0.0000	0.8496	0.8800	2.3028
5% outliers	100.0000	0.8414	0.8727	0.9578

Without L2 the clean-test accuracy changes from 0.9091 to 0.8800 (drop 0.0291); with L2 from 0.9164 to 0.8727 (drop 0.0436). The cosine similarity between the clean and contaminated weight vectors is 0.983. The shifted points clearly pull the decision boundary and change the weights, and the outliers also inflate the mean and σ used for standardization. L2 regularization did not reduce the damage here: it limits weight growth but cannot stop extreme points from tilting the boundary. Mitigations: outlier detection/removal, robust scaling (median/IQR), winsorizing or a robust loss.

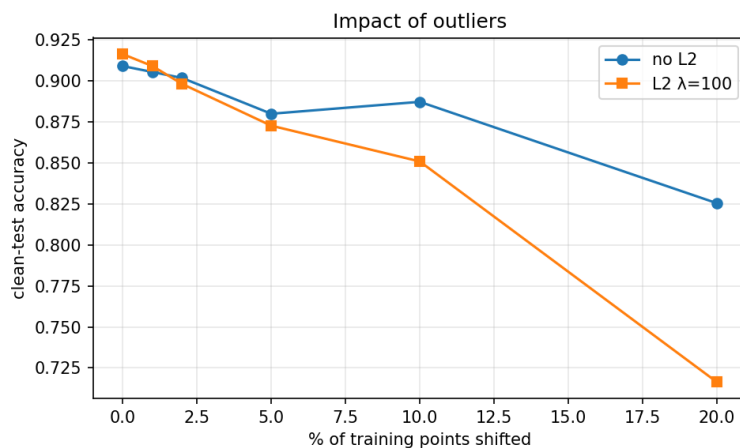


Figure 16: clean-test accuracy vs % of shifted training points

4. Conclusion

- Matrix-based linear regression is an interpretable, fast baseline (test $R^2 \approx 0.62$); closed form and gradient descent give identical predictions once features are standardized.
- L2 regularization ($\lambda^* = 3.16$) and standardization make the weights stable and comparable; the most influential features are ram, internal mem, thickness.

- Logistic regression separates genuine and forged notes with 91.6% test accuracy; L2 controls weight growth and very large λ under-fits.
- Shifting 5% of training points changed clean-test accuracy by 2.9 points (no L2) / 4.4 points (L2): outliers distort the boundary and need to be handled explicitly.